

Universal Radiative Corrections for Precision Determinations of V_{ud} BEACH conference

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- ① CKM matrix
- ② V_{us} from kaon decays
- ③ Ratio of V_{us}/V_{ud}
- ④ EFT formalism
- ⑤ V_{ud} from pion decay
- ⑥ V_{ud} from neutron decay
- ⑦ NLO EW corrections
- ⑧ NNLO EW corrections

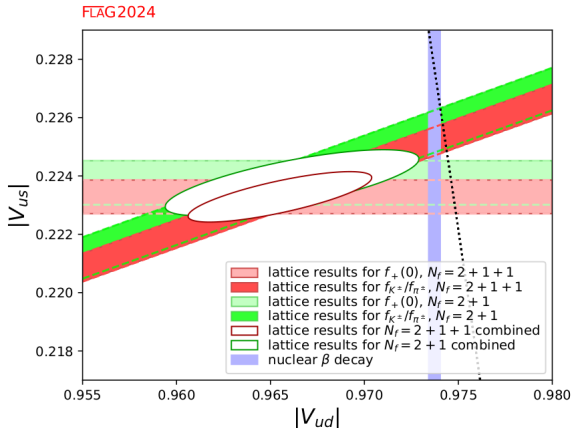
- define acronims
- more info in plots context
- smoth doen transition to my computations
- explain the calculations that have been done by martin and francesno at NNLOQCD/EW
- talk about precision of V_{ud} and is very precise with these new calculation but EW corrections can improve uncertainty which we are doing and use this to motivate our work
- add tadpole box on 2 loop
- talk about tadpoles and how they contribute
- add references

- **CKM matrix** $\longrightarrow$ quark flavour mixing in weak interactions;
- First-row unitarity test

$$|V_{ud}|^2 + |V_{us}|^2 + |V_{ub}|^2 = 1,$$

provides a sensitive probe of the SM;

- Current determinations of V_{ud} and V_{us} show a mild tension $\longrightarrow$ **Cabibbo Angle Anomaly**;
- Precision measurements of weak decays and accurate theoretical inputs are needed to clarify this tension;
- This talk:
 - Status of V_{us} from kaon decays ($K_{\ell 3}$ and $K_{\ell 2}$);
 - Status of V_{ud} from superallowed β decays, pion decay and neutron decay;
 - Ongoing work on **NNLO EW corrections for neutron decay** and V_{ud} extraction



V_{ud} and V_{us} using lattice QCD input and V_{ud} extracted from nuclear β decays.

K_{l3} decays:

$$K \rightarrow \pi \ell \nu_\ell, \quad \ell = e, \mu,$$

mediated by the weak transition

$$s \rightarrow u \ell \bar{\nu}_\ell.$$

$$\Gamma(K_{l3}) = \frac{G_F^2 M_K^5}{192\pi^3} C_K^2 S_{EW} |V_{us}|^2 |f_+^{K^0\pi^-}(0)|^2 I_{K\ell}^{(0)} \left(1 + \delta_{SU(2)}^{K\pi} + \delta_{EM}^{K\ell}\right)^2$$

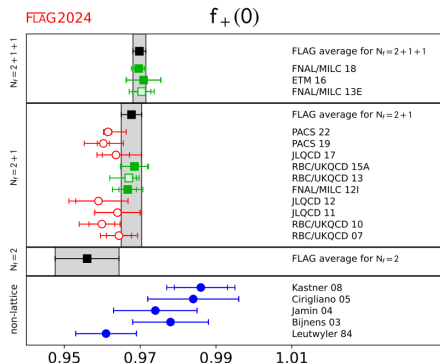
- $S_{EW} = 1.0232(3) \rightarrow$ process independent EW corrections;
- $I_{K\ell}^{(0)}$ phase space in isospin limit;
- $\delta_{SU(2)}^{K\pi}$ strong isospin breaking correction
- $\delta_{EM}^{K\ell}$ long distance EW corrections

$$\langle \pi(p_\pi) | \bar{s} \gamma^\mu u | K(p_K) \rangle = (p_K + p_\pi)^\mu f_+(q^2) + (p_K - p_\pi)^\mu f_-(q^2),$$

Experimental inputs

$$|V_{us}| f_+(0) = 0.21654(41)$$

Lattice input for $f_+(0)$



$$f_+(0) = 0.9698(17)$$

$$|V_{us}| = 0.22328(58) \quad (K_{\ell 3})$$

Lattice results (squares and empty circles) for $f_+(0)$ and ChPT estimates (filled blue circles). The black squares and grey bands indicate averages.

$$\frac{\Gamma(K_{\mu 2(\gamma)}^{\pm})}{\Gamma(\pi_{\mu 2(\gamma)}^{\pm})} = \left| \frac{V_{us}}{V_{ud}} \right|^2 \left(\frac{f_{K^{\pm}}}{f_{\pi^{\pm}}} \right)^2 \frac{M_{K^{\pm}} (1 - m_{\mu}^2/M_{K^{\pm}}^2)^2}{M_{\pi^{\pm}} (1 - m_{\mu}^2/M_{\pi^{\pm}}^2)^2} (1 + \delta_{\text{EM}})$$

$$\langle 0 | \bar{q}_1 \gamma^{\mu} \gamma_5 q_2 | P(p) \rangle = i p^{\mu} f_P, \quad P = \pi^+, K^+.$$

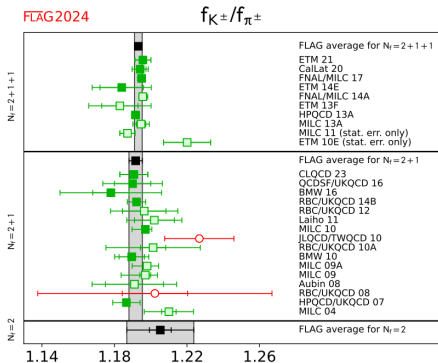
- Average of $f_K/f_p i$ in isosymmetric QCD;

Experimental inputs

$$\left| \frac{V_{us}}{V_{ud}} \right| \frac{f_{K^{\pm}}}{f_{\pi^{\pm}}} = 0.27599(41)$$

- Isospin correction from the lattice can be improved by higher statistics in contrast with extractions from ChPT;
- Isospin corrections are becoming relevant for CKM elements extraction at the current level of precision of lattice inputs;

Lattice input for f_K/f_π



Lattice results for f_K/f_π in pure QCD including isospin-breaking corrections. Black squares and grey bands indicate averages.

$$|V_{ud}| = 0.97373(31)$$

(superallowed β decays)

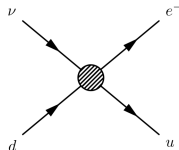
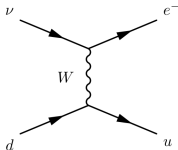
FLAG average:

$$\frac{f_K^\pm}{f_\pi^\pm} = 1.1934(19)$$

$$\left| \frac{V_{us}}{V_{ud}} \right| = 0.23126(50)$$

$$(K_{\ell 2}/\pi_{\ell 2})$$

$$H_{eff}(x) = \frac{G_F}{\sqrt{2}} V_{CKM} C(\mu) Q(x)$$



$$\frac{1}{k^2 - m_W^2} \longrightarrow \frac{-1}{m_W^2}$$

Quark decay normalized to G_F :

$$\mathcal{L}_{G_F} = -\frac{G_F}{\sqrt{2}} Q_\mu - \frac{G_F}{\sqrt{2}} V_{ud} C_q Q_q$$

$$\Gamma_{\pi\beta} = \frac{G_F^2 |V_{ud}|^2 M_\pi^5}{64\pi^3} |g_V^\pi|^2 |f_+^\pi(0)|^2 I_{\pi\ell} (1 + \Delta I_{\pi\ell})$$

- $f_+^\pi(0)$: pion vector form factor.
- $I_{\pi\ell}$: phase-space integral.
- $\Delta_{\text{RC}}^{\pi\ell}$: radiative corrections.
- g_V^π : effective vector coupling containing the short-distance electroweak corrections.

$$\Delta_{\text{RC}}^{\pi\ell} = 0.03403(11)$$

$$V_{ud}^\pi = 0.97346(283)$$

- The Behrends–Sirlin–Ademollo–Gatto theorem implies

$$f_+^\pi(0) \simeq 1 - 7 \times 10^{-6},$$

making isospin-breaking corrections negligible.

$$\frac{1}{\tau_n} = \frac{G_F^2 |V_{ud}|^2}{2\pi^3} m_e^5 f (1 + 3g_A^2) (1 + \Delta_R), \quad (1)$$

- f : phase-space factor.
- g_A : axial-vector coupling.
- Δ_R : electroweak radiative corrections.

$$\Delta_R^V = 2.436(16)\%,$$
$$|V_{ud}|_n = 0.97410(41).$$

- First NLL calculation of mixed $O(\alpha\alpha_s^2)$ corrections.
- Reduces theoretical uncertainty in Δ_R^V .
- Improves consistency of first-row CKM unitarity tests.

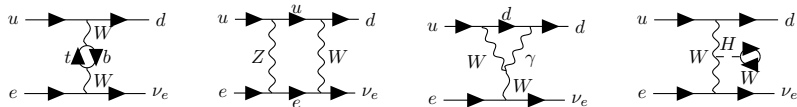


Figure 1: Representative one-loop EW diagrams contributing to the semi-leptonic process.

Wilson Coefficients at NNLO

$$C_r^{LO} = 1$$

$$C_r^{NLO} = -4 + \frac{2}{3} - 2 \log \left(\frac{\mu^2}{M_Z^2} \right) .$$

- Only box type diagrams contribute in the ratio.

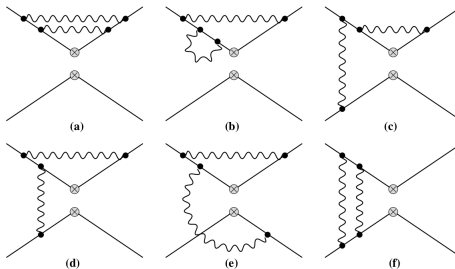


Figure 2: EFT NNLO EW corrections.

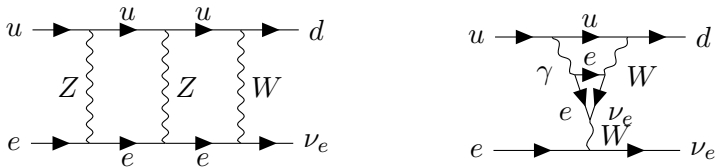


Figure 3: SM NNLO EW corrections.

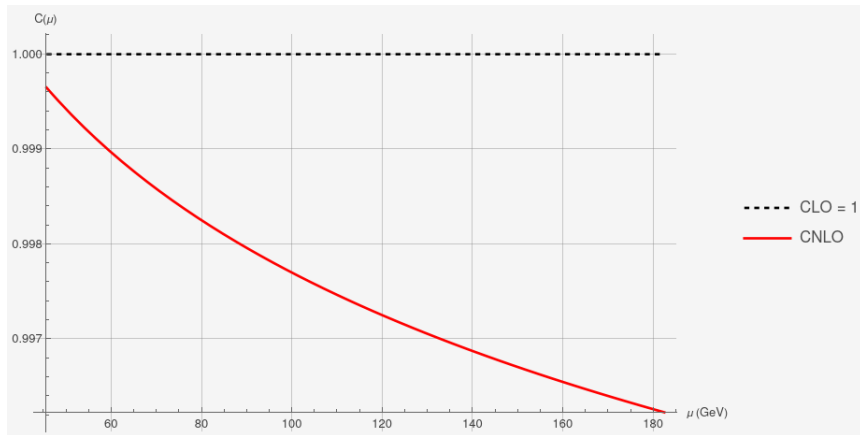


Figure 4: Evolution of the NLO WC with the scale μ .

- Combination of lattice inputs and perturbative calculations;
- Higher order calculation + sub-percent lattice calculations;
- Current ongoing work on NNLO EW corrections for V_{ud} extraction from neutron decay.